MICRO-EXPRESSION RECOGNITION BASED ON VIDEO MOTION MAGNIFICATION AND PRE-TRAINED NEURAL NETWORK

Mengjiong Bai, Roland Goecke, and Damith Herath

Human-Centred Technology, Faculty of Science and Technology, University of Canberra

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ABSTRACT

This paper investigates the effects of using video motion magnification methods based on amplitude and phase, respectively, to amplify small facial movements. We hypothesise that this approach will assist in the micro-expression recognition task. To this end, we apply the pre-trained VGGFace2 model with its excellent facial feature capturing ability to transfer learn the magnified micro-expression movement, then encode the spatial information and decode the spatial and temporal information by Bi-LSTM model. Moreover, Grad-CAM is utilised to map the model and visually explain the operating mechanism of the spatio-temporal network. Experiments on the SMIC database confirm that the proposed framework significantly improves the micro-expression recognition rate compared to without video magnification (baseline) and other state-of-the-art methods.

Index Terms— Micro-expression recognition, Motion magnification, Spatio-temporal network

1. INTRODUCTION

Micro-expressions are involuntary facial expressions with low intensity and a short duration (lasting only 1/5 to 1/25 of a second [1]). These characters make them difficult to be noticed by the naked eye. Therefore, micro-expression recognition (MER) is an interesting yet challenging problem. Many attempts have been made to detect and classify micro-expressions by artificial intelligence techniques.

Our work is inspired by Ngo *et al.* [2], who used the Eulerian video magnification method to magnify micro-expression video frames from amplitude-based and phase-based perspectives to test whether amplifying subtle facial motion, as found in micro-expressions, would boost the recognition rate. We employ the Laplacian pyramid and complex steerable pyramid, respectively, to highlight small movements, which are typical for micro-expressions. Next, we apply a pre-trained ResNet-50 network – the VGGFace2 model – to capture the spatial information for the following temporal feature learning via a Bidirectional Long Short-Term Memory (Bi-LSTM) network (Fig. 1). The phase-based approach performed best with an average accuracy of 75.76%. Our contributions are:

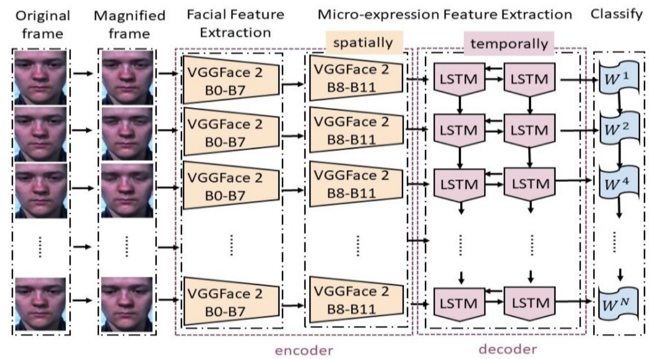

Fig. 1. Proposed framework. Amplified frames are spatially encoded by the pre-trained VGGFace2 model, before a Bi-LSTM network decodes the temporal information. *B* denotes a residual block, *W* the weight on the feature classifier layer.

1. Explore motion magnification methods on micro-expression videos and compare the performance of two methods on micro-expression recognition.
2. Propose a novel framework that combines the pre-trained VGGFace2 model and Bi-LSTM network for improved results compared to state-of-the-art methods.
3. Investigate the learning process of the spatio-temporal model's structure and use the insight gained in the model's parameter selection to improve the accuracy.

2. RELATED WORK

A few prior studies tackled the detection and classification of micro-expressions by machine learning, mainly based on hand-crafted methods but more recently also via deep learning methods. Khor *et al.* [3] used optical flow to learn the spatial relationships, plus VGG-16 and pre-trained VGGFace models to enrich the input channel and deep feature stacking. Xia *et al.* [4] proposed a model consisting of several recurrent convolutional layers to connect the spatial and temporal domains. The connectivity occurred both on pixels within the micro-expression area and optical flow. In contrast, Verburg

& Menkovski [5] only implemented RNN on the histogram of oriented optical flow features and encoded temporal movements within selected facial regions, reducing required computations and time to learn the model. Lei *et al.* [6] used a graph-temporal convolutional network (Graph-TCN) to manage the graph structures developed on the facial landmarks, extracting both node and edge features from dual channels.

Other studies focused on improving the recognition rate by amplifying the facial movements of micro-expressions. In 2012, Wu *et al.* [7] proposed Eulerian video magnification (EVM) to reveal the subtle motions in a video by decomposing the spatial information and amplifying the specific amplitude bandpass by temporal filters. This method outperforms the Lagrangian perspective algorithm. However, the error would grow quadratically with the increase of magnification factor α. To counter these issues, Wadhwa *et al.* [8] proposed a novel method to amplify the motion by computing and enhancing the phase variations in the selected temporal bandpass. This phase-based algorithm supports larger amplification factors and is notably less sensitive to noise.

Video motion amplification methods have been proposed for MER. Wang *et al.* [9] introduced an EVM with a 2^{nd} order IIR bandpass filter. Peng *et al.* [10] combined the EVM with a *Time Interpolation Model (TIM)* to reconstruct the input by magnifying the micro-expression motion and extending the temporal duration within the same model. Liu *et al.* [11] introduced an *Expression Magnification and Reduction (EMR)* domain adaptation method based on the EVM to minimise the difference between micro-expression and macro-expression. Liu *et al.* [12] combined EVM with optical flow to extract and magnify subtle movement changes for more accurate results.

3. MOTION MAGNIFICATION METHODS

The proposed framework first applies either the amplitude or phase-based motion magnification method, both inspired by the Eulerian motion magnification method [7], to the video input. Let $I(x,t)$ denote an image profile at location x and time t and let the displacement function $\delta(t)$ denote the translation motion of the image. The magnified image can be written as $I(x,t) = f(x+\delta(t))$ and $I(x,0) = f(x)$. The final goal is synthesising the signal based on the amplification factor α.

$$\hat{I}(x,t) = f((x) + (1+\alpha)\delta(t)) \tag{1}$$

3.1. Amplitude-based Motion Magnification

Here, the first-order Taylor expansion was used to approximate the image. The image intensity can be rewritten as:

$$I(x,t) \approx f(x) + \delta(t)\frac{\partial f(x)}{\partial x} \tag{2}$$

Applying a broad temporal bandpass filter $B(x,t)$ at location x and time t, $B(x,t) = \delta(t)\frac{\partial f(x)}{\partial x}$ and magnification factor α:

$$\widetilde{I}(x,t) = I(x,t) + \alpha B(x,t) \tag{3}$$

Combining Eqs. (1), (2) and (3), we obtain:

$$\widetilde{I}(x,t) \approx f(x) + (1+\alpha)\delta(t)\frac{\partial f(x)}{\partial x} \approx f(x+(1+\alpha)\delta(t)) \tag{4}$$

Let $\delta_k(t)$ denote the image intensity not entirely within the temporal filter and α_k the frequency-dependent magnification factor. Then, the final output is:

$$\widetilde{I}(x,t) \approx f(x + \sum(1+\alpha_k)\delta_k(t)) \tag{5}$$

3.2. Phase-based Motion Magnification

We use the Fourier series decomposition to approximate the motion field. The image intensity of a video frame at location x and time t can be written as:

$$f(x + (1+\alpha)\delta(t)) = \sum_{\omega=-\infty}^{\infty} A_\omega e^{i\omega(x+\delta(t))} \tag{6}$$

where each band $S_\omega(x,t)$ corresponds to the specific frequency ω, thus, $S_\omega(x,t) = A_\omega e^{i\omega(x+\delta(t))}$. The phase $\omega(x+\delta(t))$ contains the information of $S_\omega(x,t)$. Assuming it is filtered by an ideal filter and the result is $B_\omega(x,t) = \omega\delta(t)$, we can transfer the Eqs. to the sub-band with α:

$$\hat{S}_\omega(x,t) := S_\omega(x,t)e^{i\alpha B_\omega} = A_\omega e^{i\omega(x+(1+\alpha)\delta(t))} \tag{7}$$

4. MICRO-EXPRESSION RECOGNITION

4.1. Encoding Facial and Micro-expression Features

The VGGFace2 model [13] is a pre-trained network based on the ResNet-50 model and the VGGFace2 database, which resulted in outstanding performance in facial recognition tasks. In the proposed framework, this model is utilised to extract the facial features. Fig. 2 illustrates the model's structure.

The ResNet-50 residual learning framework [14] consists of residual blocks (each containing three convolutional layers), which could solve the degradation problem and keep the details from shallower layers. Initially, we employed the Grad-CAM technique [15] to visualise the activation maps of each residual block. We observed that the block with activation regions closest to the facial features was block B_8. Hence, we froze the blocks before B_9, while retraining blocks $\{B_9 - B_{16}\}$ on the micro-expression database (see Fig. 2).

4.2. Micro-expression Feature Decoder

Due to the low intensity and short duration, micro-expression features are extracted at both the spatial and temporal level to obtain more completed motion information. To capture the

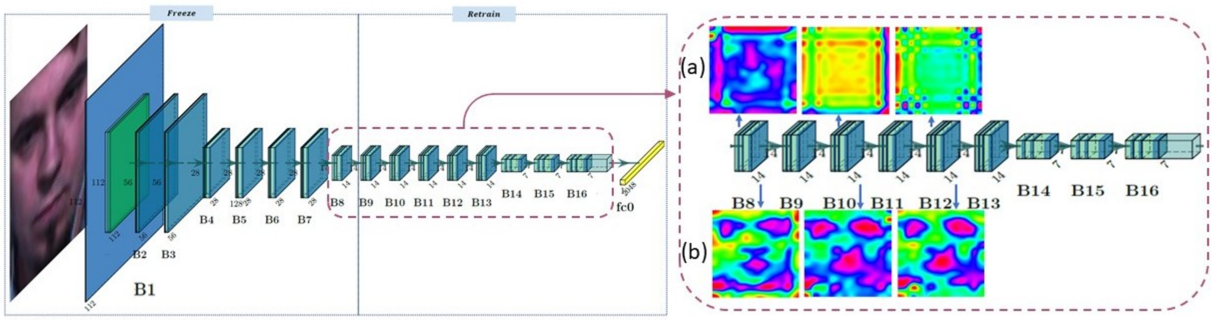

Fig. 2. VGGFace2 model structure and Grad-CAM computed activation heat maps on residual blocks B_8, B_{10}, B_{12}. (a) Before retraining, highlighted regions focus on the face only in the block B_8, while spreading across the entire frame in B_{10} and B_{12}. (b) After retraining, purple highlighted regions gradually focus on the micro-expression related areas.

micro-expression features from sequential samples, we employed a Bi-LSTM network, which is an extension of traditional LSTMs, to access long-range dynamic information by considering both backward and forward activation [16].

5. EXPERIMENTS

5.1. SMIC Database

The spontaneous micro-expression corpus (SMIC) is a common MER benchmark dataset, which contains High Speed (HS) micro-expression samples recorded at 100fps. In the dataset, the shortest recorded expression was 11 frames long (~ 0.11s), and the average expression length was 29 frames (~ 0.30s). In total, there are 164 videos in the SMIC HS, 70 for negative, 51 for positive and 43 for surprise [17].

5.2. Micro-expression Motion Magnification

Amplitude-based Motion Magnification (AMM) consists of four steps: first, each frame is decomposed into different spatial frequency bands by a full Laplacian pyramid. Next, a Butterworth filter extracts the frequency bands of interest. Then, the magnification factor α is multiplied with different bandpassed signals and the amplified signal is added to the original image. Wu *et al.* [7] state that the frequency range should be equal to the micro-expressions duration time, i.e. $\frac{1}{5}$ to $\frac{1}{25}$, but as applying these values caused noticeable noise, we narrowed the boundaries to $\frac{1}{15}$ to $\frac{1}{25}$ in line with [2].

Phase-based Motion Magnification (PMM) applies octave complex steerable pyramids to decompose the image of local phase over time with four different orientations, then followed with temporal filtering. From [8], steerable pyramid enables the computing process better noise reduction ability because it has impulse response with spatial support. So it's easier to isolate the specific frequencies while reducing the force of the rest of the components. Our experiments have confirmed this and reflected it in the Fig. 3.

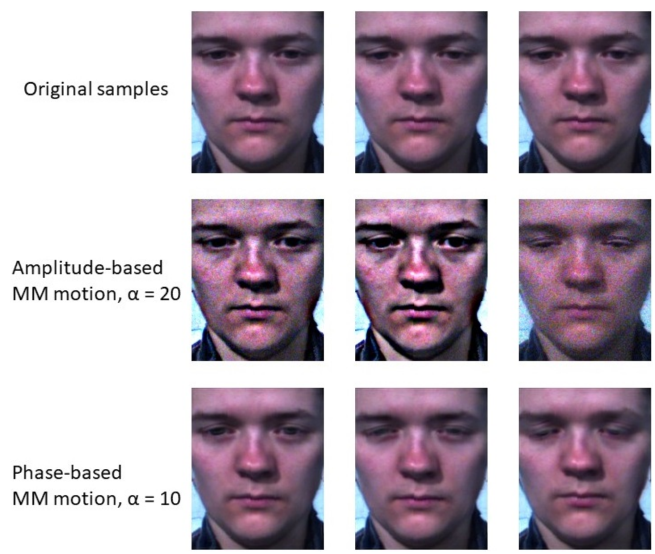

Fig. 3. Example video frames of amplified facial movements by two motion magnification methods

5.3. Micro-expression Recognition

5.3.1. Data Preprocessing

The experiments were conducted without face alignment and face cropping to avoid potential errors. Video frames were resized to a resolution of 224×224 pixels. We augmented the training data by extracting fixed-length video segments (sequences of consecutive frames) from each original video by overlapping sliding windows of size N and a step size of 1. As small N leads to a loss of long-term dependence information, while large N reduces the number of sequences, we evaluated several combinations to find a suitable value for N.

As shown in Table 1, the best accuracy by both amplitude and phase based methods was achieved for $N = 14$. The best result mostly appeared in the *positive* micro-expression. We hypothesise that as 14 is around half of the average video du-

	N	7	8	9	10	11	12	13	14
AMM	**Avg.**	**0.67**	**0.67**	**0.74**	**0.68**	**0.68**	**0.64**	**0.62**	**0.73**
	Neg	0.54	0.70	0.61	0.57	0.73	0.63	0.64	0.72
	Pos	0.70	0.70	0.78	0.62	0.75	0.71	0.50	0.80
	Sur	0.80	0.42	0.56	0.50	0.60	0.60	0.63	0.70
PMM	**Avg.**	**0.70**	**0.70**	**0.76**	**0.71**	**0.73**	**0.70**	**0.67**	**0.79**
	Neg	0.67	0.71	0.65	0.75	0.68	0.63	0.65	0.74
	Pos	0.75	0.71	0.75	0.67	0.86	0.60	0.80	0.75
	Sur	0.73	0.67	0.67	0.71	0.71	0.67	0.64	1.00

Table 1. Experimental results on SMIC [17]: Average prediction accuracy and accuracy for each micro-expression class (*Neg*: negative, *Pos*: positive, *Sur*: surprise) for different sliding window sizes N. *AMM*: amplitude-based motion magnification, *PMM*: phase-based motion magnification.

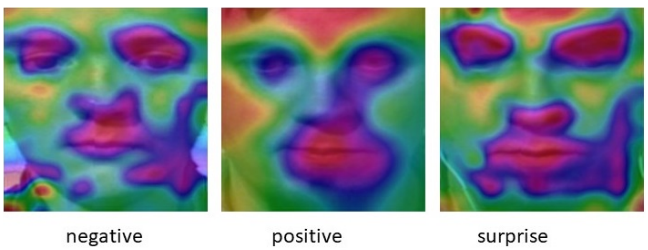

Fig. 4. Activation heat maps. (*Red highlight*) Different face regions correspond to different micro-expressions. For *negative*, it focuses more on the eyes; for *positive*, on the mouth corners; and for *surprise*, more on the eyebrows.

ration, it is easier to capture the process of micro-expressions. In the subsequent experiment, we chose $N = 14$ to conduct a 5-fold cross-validation on the proposed framework.

5.3.2. *Model Training and Parameter Selection*

Initially, we load the ResNet-50 network with the VGGFace2 model weights. As Section 4.1 detailed, the blocks' parameters before B_8 are frozen, while the remaining blocks participate in the training. Two linear layers were added after the last residual blocks to produce the feature vector following sequential learning. For the decoding part, there are two hidden layers and 512 hidden nodes. One softmax and one ReLU layer were applied at the end of the network

We hypothesise that blocks *before* B_9 primarily extract facial geometry features, while the remaining blocks learn micro-expression features. Fig. 4 shows that the blocks *after* B_8 capture the corresponding characteristics of different micro-expressions (activation focus in different face areas).

5.4. Results and Discussion

We utilised 20% data for testing; totally, there are 33 videos of variable length – overlapping sliding window is not applied

Source paper	Method	Year	Accuracy
[19]	3D-FCNN	2018	55.49%
[20]	Hierarchical scheme	2018	66.46%
[21]	MicroExpSTCNN	2019	68.75%
[10]	EVM+TIM	2019	68.90%
[18](baseline)	VGGFace2+Bi-LSTM	2020	60.60%
[22]	SETFNet	2020	70.25%
Proposed	**EVM+VGGFace2+Bi-LSTM**	**2021**	**75.76%**

Table 2. Result comparison in terms of the best accuracy of thd proposed method with state-of-the-art methods. The results are taken from the respective source papers.

during testing. In comparison to the state-of-the-art, the proposed EVM + VGGFace2 + Bi-LSTM framework achieved a significantly improved average accuracy of 75.76% (Table 2).

In terms of the individual contributions of the components of the framework, we compared our experimental results to previous studies on the SMIC dataset [18] (baseline), which also used a VGGFace2 + Bi-LSTM pipeline. We can see a performance improvement from 60.60% to 75.76%.

Furthermore, we tested replacing the VGGFace2 model with a ResNet-50 model pre-trained on ImageNet, which achieved an average accuracy of 37.9%. Using only the VGGFace2 pre-trained model with a fully-connected layer before classification achieved an average accuracy of 41.9%. Although the improvement is not large, because it is hard to learn the micro-expression features solely relying on spatial information, it shows the advantage of using a model pre-trained on face images. We hypothesise that such a model is better capable of extracting subtle facial deformations, the detection of which is crucial for MER.

Positive micro-expressions were more often correctly recognised than negative and surprise micro-expressions, which may be due to a slightly longer duration of the positive micro-expressions. Further work is required to confirm this.

6. CONCLUSIONS

The Eulerian motion magnification exaggerated the micro-expression deformation at spatial and temporal level and indeed increased the recognition rate. The phase-based method outperformed the amplitude-based method, with the highest average accuracy of 75.76%. Moreover, the VGGface2 model provides effective assistance on extracting the spatial features and enables the proposed framework to accurately capture the important regions of different kinds of micro-expressions. However, both amplitude and phase-based motion magnification require manual parameter fine-tuning, such as the temporal frequency boundary and the magnification factor.

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